

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 27)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0	1
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0	1
---	---

Explain **three** problems this spoil heap could cause and describe how each could be managed.

[6 marks]

0	1
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Explain why deep layers of soil are applied when a spoil heap containing toxic materials is restored.

[2 marks]

0	1
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Explain why an old steel-works spoil heap may be valuable for wildlife conservation.

[2 marks]

10

02 . 1

Describe how hydrothermal deposition produces exploitable concentrations of metal ores.

[4 marks]

02 . 2

Explain how Proterozoic marine sediments produced the world's major iron ore deposits.

[2 marks]

02 . 3

Explain how alluvial (physical sediment) processes separate and concentrate minerals such as gold and tin ore.

[2 marks]

02 . 4

Give **two** examples of mineral deposits formed by biological sedimentation and state their origin.

[2 marks]

1

2

10

0 3 Table 1 lists some techniques used in ecological monitoring.

0 3 . **1** Complete Table 1 by giving one application and one limitation of each technique.

[6 marks]

Table 1

Technique	Application	Limitation	
Image databases of unique markings			
eDNA sampling			
Geolocator tags			

03 . 2

Explain how satellite radar (microwave) sensors are useful for monitoring oil pollution and sea conditions.

[2 marks]

03 . 3

Give **two** types of indirect evidence of an animal's presence and the information each can give.

[2 marks]

1

2

10

04

Table 1 shows the rate of heat loss through 1 m^2 of different glazing systems for a 1°C temperature difference (U-value).

Table 1

Glazing	Heat loss / $\text{W m}^{-2} \text{ K}^{-1}$
Single glazing	5.8
Double glazing, air filled	2.8
Double glazing, argon filled, low-e glass	1.6
Triple glazing, krypton filled, low-e glass	0.8
Double glazing, vacuum (0.2 mm gap)	0.74

04 . 1

A house has 20 m^2 of single glazing. The average temperature difference between inside and outside over the heating season is 12°C . Calculate the reduction in the rate of heat loss if the windows are replaced with argon-filled low-e double glazing.

[3 marks]

Answer

W

0 4 . 2

Explain why the gap between the panes should be as large as possible but not too large.

[2 marks]

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0 4 . 3

Explain how low-emissivity glass and inert gas filling each reduce heat loss.

[2 marks]

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0 4 . 4

Explain why vacuum double glazing has a very narrow gap and where it is used.

[2 marks]

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0 4 . 5

State **one** reason why insulation thickness has an optimum value.

[1 mark]

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10

0 5 A nuclear reprocessing plant on the coast discharges low-level liquid radioactive waste into the sea through a pipeline.

0 5 . **1** Explain what is meant by **Critical Pathway Analysis** (CPA).

[2 marks]

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0 5 . **2** Give **three** factors that would be considered in a CPA for this discharge.

[3 marks]

1

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2

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3

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0 5 . **3** Describe how **Critical Group Monitoring** (CGM) would be used to assess the risk to the public from this discharge.

[3 marks]

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0 5 . **4** Suggest **two** types of environmental sample that would be collected to check the CPA predictions.

[2 marks]

1

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2

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10

0 6 . 1 Explain how each of the following human activities alters processes in the hydrological cycle.

[6 marks]

Deforestation

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Urban development

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Agriculture

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0 6 . 2 Describe **two** ways in which global climate change alters the hydrological cycle.

[2 marks]

1

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2

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0 6 . 3 Explain why the impact of industrial and domestic water use on the hydrological cycle is usually local.

[2 marks]

10

07 . 1

Define the term **persistence** as applied to a pollutant.

[1 mark]

07 . 2

Explain how the persistence of a pesticide is affected by **two** named environmental factors.

[4 marks]

5

08 . 1

Which of the following is the main reason that lead-based paint is a greater risk to children than to adults? Tick one box.

[1 mark]

- A Children inhale more lead vapour
- B Children pick up dust / flakes and suck their fingers
- C Lead is more soluble in children's blood
- D Children spend more time near roads

☐☐☐☐

08 . 2

Explain how tri-butyl tin (TBT) in anti-fouling paint caused pollution even though only small amounts dissolved into the sea.

[3 marks]

08 . 3

Iron is not toxic, yet iron-rich drainage from mine spoil heaps kills fish. Explain why, and describe how it is controlled.

[3 marks]

08 . 4

Give **two** uses of lead that continue because they do not cause significant lead release into the environment.

[2 marks]

1

2

08 . 5

Explain why heavy metals are described as neurotoxins.

[1 mark]

10

09 . 1

Explain how each of the following environmental factors affects the rate of degradation of pollutants. Give a named example of a pollutant for each factor.

[6 marks]

Temperature

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Light

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Oxygen

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09 . 2

Explain how the presence of adsorbent materials such as clay can reduce the severity of pollution by heavy metals or radioactive caesium-137.

[2 marks]

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09 . 3

Explain why a fall in pH increases the harm caused by many heavy metals.

[2 marks]

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10

$$\boxed{1} \boxed{0} . \boxed{1}$$

Evaluate the potential contribution of renewable energy resources to future energy supplies in a country such as the UK.

[9 marks]

Extra space

10 . 2

State the factor by which the kinetic energy of wind increases when wind velocity doubles.

[1 mark]

10

Do not write outside the box

Write an essay on one of the following topics.

1	1	.	1
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Discuss how an understanding of the hydrological cycle can be used to develop strategies for the sustainable management of water resources.

[25 marks]

OR

1 1 . 2

Discuss how knowledge of the properties of pesticides can be used to reduce their environmental impacts while maintaining food production.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

[illegible]

END OF QUESTIONS